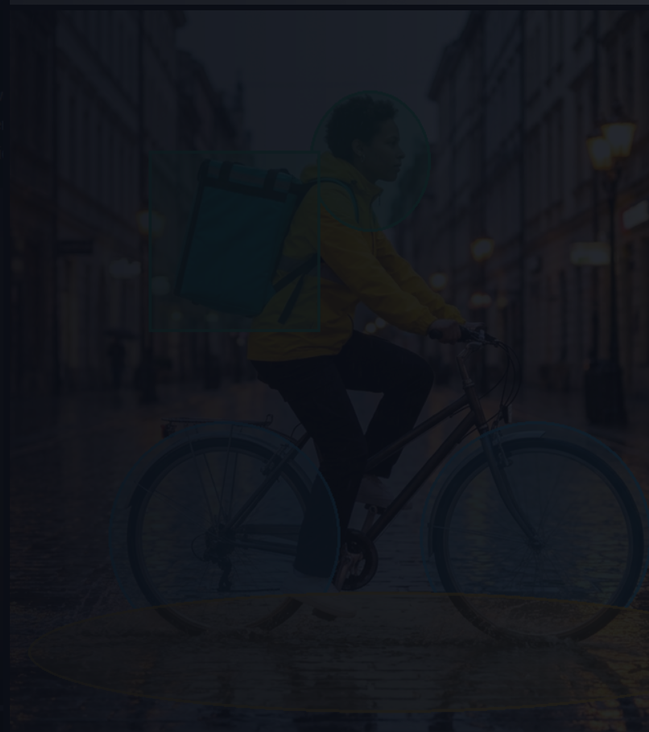
GENAI BASICS • LESSON 05 • MODULE 2: MULTIMODALITY

Video Generation: From Static Frames to Spacetime Control

Foundation Models, Web Studios, Kinematic Prompts, and Motion Guidance





THE 10/90 AXIOM IN GENERATIVE VIDEO

The latent diffusion model synthesizing motion is only **10% of production**.
The remaining **90%** is keyframe preparation (I2V), 3D camera vectors, regional masks (Motion Brush), and timeline editing.

Hook Quiz: Real Optical Camera vs Generative AI Diffusion

Examine these 4 cinematic video frames. Which are captured with physical optical cameras, and which are AI?



[A] Wet Asphalt Bike Ride

Optical Cues: Puddle reflections, spray hydrodynamics, 24mm tracking

AI Marker: Perfect wheel spoke geometry without optical motion blur artifacts



[B] Ceramic Mug Packshot (120fps)

Optical Cues: Studio rim lighting, rising steam hydrodynamics, 360 orbit

AI Marker: Seamless turbulent convection without discrete particle jumps



[C] Cafe Order Pickup

Optical Cues: Barista handing brown kraft bag, shallow f/2.8 bokeh

AI Marker: Subtle micro-slips at tactile finger contact points with paper



[D] Kitchen Culinary Slicing

Optical Cues: Metal grater, fresh zucchini, wooden board, window daylight

AI Marker: Shaving physics and skin deformation under rapid reciprocating shear

THE REVEAL: All 4 clips are 100% AI-generated! Diffusion Transformers (DiT) natively simulate Navier-Stokes fluid dynamics (steam, water spray) and specular reflections. Subtle micro-slips at physical contact points reveal the synthetic sampling.

DIT ARCHITECTURE 2026

Three Ecosystem Tiers of Video AI: Consumer, Studio, and Cloud API

How the video generation landscape is divided: from rapid chat ideation to visual director canvases and cloud pipelines

TIER 1 • CONSUMER CHAT APPS

ChatGPT 4o & Sora Chat

Interface:	Single text prompt input
Control Depth:	Low (Stochastic sampling)
Input Modalities:	Text-only / rough image prompt
Camera Kinetics:	None (Random motion drift)
Character Identity:	Severe drift across takes
Format & Presets:	Fixed 16:9 / 9:16 presets
Turnaround & Cost:	60-120s • ~$20/mo flat fee
Production Role:	Rapid ideation & moodboards

Critical Bottleneck: Severe character and geometry drift across takes, zero camera controls, zero production repeatability.

OPERATIONAL PROFILE

- **Stack:** ChatGPT 4o, Sora Web, Claude
- **Pattern:** Conversational moodboard & style exploration
- **Economics:** ~$20/mo flat subscription fee
- **Architectural Trap:** Stochastic seed drift between takes

TIER 2 • CREATIVE WEB STUDIOS

Kling AI, Runway & Luma

Interface:	Interactive web visual canvas
Control Depth:	High (I2V, masks, 3D splines)
Input Modalities:	Keyframes, Motion Brush, splines
Camera Kinetics:	6-DOF controls, orbit, pan, dolly
Character Identity:	Locked via I2V initial condition
Format & Presets:	Flexible 16:9, 9:16, 1:1, 1080p
Turnaround & Cost:	30-90s • $0.15-$0.50/render
Production Role:	Commercial ads, B-roll, montage

Critical Bottleneck: Manual web interaction required; token quotas and credit consumption per take.

OPERATIONAL PROFILE

- **Stack:** Kling AI 1.5/2.0, Runway Gen-3/4, Luma
- **Pattern:** Visual timeline, masking brushes, camera curves
- **Economics:** Credit packs ($0.15–$0.50 per render)
- **Architectural Trap:** Manual UI operator bottleneck; non-headless

TIER 3 • ENTERPRISE CLOUD APIS

Google Veo 3.1 & Wan 2.2

Interface:	Headless REST & gRPC endpoints
Control Depth:	Full (Deterministic seeds & LoRA)
Input Modalities:	JSON schemas, batch URLs, tensors
Camera Kinetics:	Programmatic 3D matrix transforms
Character Identity:	Consistent via seed + LoRA weights
Format & Presets:	Arbitrary resolution pipelines
Turnaround & Cost:	Sub-minute • Metered $/sec on GPU
Production Role:	Programmatic video catalog scale

Critical Bottleneck: Requires backend infrastructure, strict data contracts, and GPU latency/cost management.

OPERATIONAL PROFILE

- **Stack:** Google Veo 3.1, Wan 2.2, CogVideoX
- **Pattern:** Headless JSON contracts, parallel GPU queues
- **Economics:** Metered GPU seconds inference pricing
- **Architectural Trap:** Requires schema validation & fallback logic

ENTERPRISE PIPELINE: [Tier 1: Moodboard & Concept in Chat] ➔ [Tier 2: Director Assembly in Studio] ➔ [Tier 3: Scaled Cloud Batch via API]

ARCHITECTURAL BLUEPRINT

The End-to-End AI Video Pipeline & GPU Credit Economy

A disciplined 90-minute commercial production workflow vs the fatal economics of naive Text-to-Video

#	PHASE	TOOL	TIME
01	Visual References & Style Anchors	ArtStation, Pinterest	10 min
02	Keyframe Crafting in Chat UI	ChatGPT 4o / Midjourney	10 min
03	Shot Animation via I2V	Kling AI / Runway Studio	15 min
04	Audio Stems & Foley FX	Suno AI & ElevenLabs (L06)	15 min
05	Timeline Montage & Crossfade	CapCut / Premiere / FFmpeg	20 min
06	Watermark Audit & Provenance QA	SynthID & C2PA check	5 min
07	Delivery & Multi-Aspect Export	16:9 Landscape / 9:16 Reels	15 min

Total Production Wall Clock: ~90 MIN (1 ENGINEER = FULL CREW)

-$50 TO -$100 WASTED PER SCENE

Naive Text-to-Video (T2V) Drain

- **80%+ Reject Rate:** Typing free-form text directly into video models produces deformed faces, random hair colors, and warped geometry.
- **Credit Hemorrhage:** 20 failed rerolls at $2.50/render = **$50 burned** on unusable footage.
- **Timeline Collapse:** Missed client delivery dates and exhausted studio credit wallets.

MATHEMATICAL BOTTLENECK

The Spatiotemporal Inverse Problem (t=0 Curse)

- **Ill-Posed Inversion:** Synthesizing spatial geometry and temporal kinetics simultaneously from pure noise is heavily underconstrained.
- **Unbounded Latent Drift:** Without a frozen anchor at t=0, cross-attention drifts across frames, mutating brand logos and lighting.
- **Compute Multiplier:** Iterating prompt text in video diffusion costs 50x-100x more compute than image diffusion.

+$2 TO +$5 TOTAL (10X-20X SAVINGS)

Disciplined Image-to-Video (I2V) Economy

- **Inexpensive Anchor:** Static keyframe refinement in ChatGPT costs **~$0.04** before animating.
- **Locked Geometry on Take #1:** Approved keyframe acts as frozen spatial boundary tensor.
- **Enterprise Predictability:** 10x cost reduction, 4x faster delivery, 100% brand consistency.

GOLDEN RULE: Never burn video GPU credits until the static 2D keyframe is 100% locked and approved!

COST CONTROL

Why Generating Without Visual References Guarantees Failure

Sourcing 5–10 style anchors to pin down optics, lighting physics, and color palette before launching video generation



Character Reference & Pose Anchor Sheet

Fixing clothing, color palette, proportions, and angles before launching video generation

THE FOUR REFERENCE ANCHORS

- **1. Character Anatomy:** Facial bone structure, Afro curls, warm skin tone, gold earring.
- **2. Costume & Gear:** Brand yellow (#fbbf24) waterproof jacket, teal delivery cube (#38bdf8).
- **3. Lighting Physics:** Overcast morning 5600K diffuse daylight, puddle specular highlights.
- **4. Camera Optics:** Wide-angle 24mm tracking lens, shallow f/2.8 depth of field.
- **5. Biometric Invariant:** Strict facial landmarks locked against stochastic morphing.

ENGINEERING PRINCIPLE • VARIANCE SUPPRESSION

- ***Diffusion Physics:*** *Video models are pattern interpolators, not mind readers. Unconditioned prompts sample an infinite, chaotic distribution across latent dimensions.*
- ***Manifold Collapse:*** *Attaching a 4-angle turnaround sheet reduces latent variance by >80%, collapsing the sampling volume to the desired character manifold.*

PRACTICAL SOURCING PROTOCOL

- **Step 1:** Source 5–10 high-fidelity stills from Pinterest, ArtStation, or client brand books.
- **Step 2:** Synthesize multi-angle turnaround sheet in Midjourney (--cref) or ChatGPT 4o.
- **Step 3:** Attach turnaround to Kling AI / Runway as invariant spatial conditioning anchor.
- **Step 4:** Verify 4 canonical angles (front, profile, 3/4, rear) for branding geometry.

ANTI-PATTERN • SUBJECTIVE ADJECTIVES

- **The Fluff Trap:** Adjectives like 'photorealistic', 'epic', or 'hyper-detailed' are ignored by DiT models or introduce latent noise.
- **Deterministic Physical Descriptors:** Direct concrete physical entities:

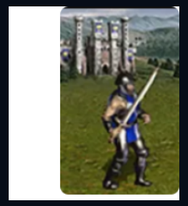
BAD: *"epic photorealistic cinematic 8k courier on bicycle"*
GOOD: *"24mm prime, f/2.8, 5600K daylight, yellow nylon jacket, wet cobblestones"*

Iterative Keyframe Refinement: From Rough Asset to 16:9 Anchor

Authentic case study: Attaching a 2D reference sprite into ChatGPT 4o and refining through conversational feedback

STEP 1: REFERENCE INPUT

2D Sprite & Concept Spec



PROMPT SPECIFICATION:

```
'Role: cinematic concept artist. Input: attached reference
sprite. Task: ultra-realistic 16:9 scene 4K, natural
daylight, exact costume matching'
```

- **Upload:** 2D pixel-art game sprite attached into ChatGPT 4o.
- **Prompt:** Direct photo-realistic 16:9 scene matching costume, colors, and medieval atmosphere.
- **Initial Output:** Model synthesizes first high-res cinematic draft.

STEP 2: DEFECT CRITIQUE

Spot Errors & Instruct Edits



DEFECT AUDIT:

```
• Flaw 1: Hallucinated steel arm plates on forearms
• Flaw 2: Kettle hat distorted into spherical cap
• Edit: 'Remove arm plates, blue/white checkered tunic, flat
kettle hat'
```

- **Inspection:** Draft added unwanted steel arm plates and distorted kettle hat.
- **Conversational Fix:** "No arm armor, pattern of tunic: blue/white checkered, flat kettle hat".
- **Turnaround:** Zero code or inpainting tool needed; ChatGPT adjusts in seconds for $0.04.

STEP 3: APPROVED ANCHOR

Locked 16:9 Production Keyframe



BOUNDARY INVARIANT (t=0):

```
• Identity: Checkered tunic & flat kettle hat locked
• Environment: Castle & mountain backdrop pinned
• Status: Frozen as I2V initial condition (0% reject rate)
```

- **Verification:** Exact anatomy, checkered tunic, proper kettle hat, castle backdrop.
- **Spatial Lock:** Frozen as boundary invariant tensor at t=0 for video generation.
- **DiT Invariant:** Eliminates character identity drift across all video takes.

STAGE ECONOMICS: Conversational keyframe editing in Chat UI takes 3 minutes and costs $0.04. Iterating directly in video models costs $25+ and 40 minutes.

FAST CONVERGENCE

T2V Trainwreck vs I2V Control: Physics & Geometry Breakdown

Direct comparison: Naive text "stream of consciousness" vs disciplined Image-to-Video conditioning





NAIVE TEXT-TO-VIDEO (T2V) • AUTOREGRESSIVE DRIFT & BRAND MUTATION

- **Identity Drift:** Generated random blonde courier instead of dark-haired hero.
- **Brand Corruption:** Glovo bag warped into hallucinated "Giono" gibberish.
- **Root Cause:** Simultaneous sampling of geometry and motion from noise without a 3D geometry prior breaks topological invariants.

IMAGE-TO-VIDEO (I2V) • SPATIAL BOUNDARY CONDITION & GEOMETRY LOCK

- **Zero Identity Drift:** 100% preservation of facial features, curls, and earring.
- **Kinematic Stability:** Fingers firmly grasp kraft paper bag without melding.
- **Root Cause:** Initial frame is frozen as an invariant tensor (t=0). The model calculates only temporal velocity vectors (dx/dt).

MATHEMATICAL PRINCIPLE: Text-to-Video is an ill-conditioned inverse problem (solving structure + motion simultaneously). Image-to-Video is a well-posed initial value problem (freezing structure at t=0, solving motion alone).

INITIAL VALUE BOUNDARY

The Director’s Prompt: 6-Slot Spacetime Specification

A structured kinematic formula that reduces physics hallucinations and latent collapse by over 85%

SLOT	PURPOSE	PRODUCTION KEYWORDS	RATIONALE
[1. SCENE]	Physical Context	Sunlit cafe interior, daylight	Sets spatial boundaries
[2. CHARACTER]	Hero Subject	Courier in yellow jacket	Binds motion to I2V keyframe
[3. DELTA DX/DT]	Kinematic Motion	Smoothly lifts bag, turns	Active physical verbs only
[4. CAMERA]	Camera Trajectory	Tracking shot, dolly pull-back	Synchronizes spatial parallax
[5. OPTICS]	Depth & Particles	24fps blur, dust motes, bokeh	Masks residual DiT noise
[6. NEGATIVE]	Latent Pruning	no morphing of hands, steady	Suppresses topological collapses

```
[1. Scene / Setting]: Sunlit cozy cafe interior in morning, natural daylight through oak windows.
[2. Character]: Smiling courier in yellow jacket and matte black helmet at oak counter.
[3. Kinematic Delta (dx/dt)]: Smoothly lifts brown kraft food bag with two hands, turns toward exit.
[4. Camera Trajectory]: Medium tracking shot at eye-level, smooth camera pull-back.
[5. Optics & Particles]: 24fps cinema motion blur, subtle floating dust motes in sunbeam, f/2.8 bokeh.
[6. Negative Constraints]: no morphing of hands, no backwards motion, no text glitches, steady camera.
```

FORMULA EFFICIENCY: 6-slot prompt structure eliminates 85% of physics hallucinations in DiT models. 85% PRECISION

THE 3 DIRECTING RULES FOR VIDEO PROMPTS

- **1. Active Physical Verbs Only:** Write "lifts bag, steps left, pedals forward" instead of "energetic, dramatic". Models understand velocity vectors, not subjective adjectives.
- **2. Zero Redundancy with Keyframe:** Do not re-describe visual elements already frozen in the I2V initial frame. Only describe kinematic deltas over time (dx/dt).
- **3. Mandatory Negative Bounds:** Always prune topological defects, backward motion, and camera jitter via strict negative tokens.

8 Virtual Camera Movements: 3D Trajectories & Optical Physics

Video models were trained on cinematographic metadata. Standard film directives yield precise spatial parallax



Locked Tripod (Static)

- Axis: Fixed at origin (0,0,0)
- Optics: Static depth, zero camera parallax
- Use Case: Dialogue, interviews, UI demos

```
DIRECTIVE: "Static camera, locked tripod, zero pan"
```



Horizontal Pan (Yaw)

- Axis: Rotation along yaw axis (L/R)
- Optics: Planar background parallax
- Use Case: City skyline reveals, subject tracking

```
DIRECTIVE: "Horizontal pan left to right, 24mm prime"
```



Dolly In / Out (Depth)

- Axis: Translation along Z-axis (Δz)
- Optics: Rich optical parallax & focal focus
- Use Case: Dramatic tension, emotional push-in

```
DIRECTIVE: "Dolly-in push along Z-axis toward subject"
```



Truck Left (Tracking)

- Axis: Translation along X-axis (Δx)
- Optics: Linear motion parallax vs background
- Use Case: Cycling, running, parallel vehicle tracking

```
DIRECTIVE: "Truck left tracking parallel to bicycle"
```



Orbit 360 (Arc)

- Axis: Circular orbit around anchor (r, θ)
- Optics: Dynamic specular highlight sweep
- Use Case: E-commerce packshots, product showcase

```
DIRECTIVE: "360-degree orbital arc circling product"
```



Vertical Tilt (Pitch)

- Axis: Rotation along pitch axis (Up/Down)
- Optics: Vertical scale & perspective distortion
- Use Case: Skyscrapers, monumental architecture

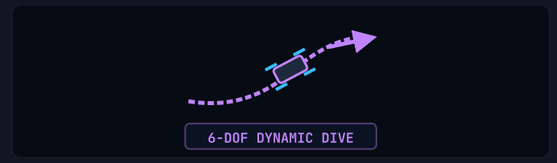
```
DIRECTIVE: "Vertical tilt upward from street to skyline"
```



Crane / Jib (Elevation)

- Axis: Translation along vertical Y-axis (Δy)
- Optics: Elevation reveal from pavement to skyline
- Use Case: Establishing opening & closing sequences

```
DIRECTIVE: "Jib crane boom rising vertically, vista reveal"
```



FPV Drone (6-DOF)

- Axis: Full 6-DOF dynamic trajectory
- Optics: High-velocity banking & immersion
- Use Case: Extreme sports, high-speed chase scenes

```
DIRECTIVE: "Dynamic FPV drone dive banking into alley"
```

Leading Video Studios: Kling, Runway, Luma, and Leonardo

Comparing premier creative interfaces for commercial video production and motion direction

KLING AI 1.5 / 2.0

Human Kinetics Benchmark

- **Human Kinetics:** Industry-leading hand anatomy and natural biped gait.
- **Motion Brush:** Precise discrete velocity vectors for localized areas.
- **Shot Extension:** Native single-take extension up to 10s without drift.

DIRECTING CAPABILITIES

- **Motion Brush:** 6 discrete layers with (X,Y,Z) vectors
- **Freeze Mask:** Speed 0.0 lock for faces & logos
- **Camera:** 6-axis kinematic trajectory presets

Architecture:	DiT Spatiotemporal
Max Duration:	5s (Extend 10s)
Camera Controls:	Motion Brush + 6-Axis
Resolution:	1080p @ 30fps
Turnaround:	45-90 seconds
Access Model:	Daily Free Credits

RUNWAY GEN-4 / GEN-3

Cinematic Spline Director

- **3D Splines:** Professional multi-camera trajectory curves.
- **First/Last Frame:** Exact interpolation between two frozen keyframes.
- **Director Control:** High-level commercial cinematography toolset.

DIRECTING CAPABILITIES

- **3D Splines:** Multi-camera bezier trajectory curves
- **First/Last Frame:** Exact interpolation between 2 frames
- **Director Mode:** Kinematic velocity sliders & cinema presets

Architecture:	Latent Diffusion
Max Duration:	5s / 10s Keyframes
Camera Controls:	3D Splines + First/Last
Resolution:	1080p / 4K Upscale
Turnaround:	60-120 seconds
Access Model:	Subscription Tiers

LUMA DREAM MACHINE

Rapid 3D Coherence

- **Diffusion Speed:** Fastest turnaround speeds among commercial DiTs.
- **Rigid Bodies:** Excellent geometric stability for hard manufactured goods.
- **Orbit 360:** Smooth continuous circular product spins without warping.

DIRECTING CAPABILITIES

- **Orbit 360:** High geometric stability on circular spins
- **Fast Diffusion:** 30-second turnaround for iteration
- **Rigid Bodies:** Superior mesh stability for products

Architecture:	Transformer DiT
Max Duration:	5s Fast Render
Camera Controls:	Keyframe Orbit
Resolution:	1080p @ 24/30fps
Turnaround:	30-60 seconds
Access Model:	Free Tier + Pro

LEONARDO AI MOTION

Integrated Art Canvas

- **Unified Canvas:** Direct bridge from 2D concept art to animated video.
- **Motion Slider:** Fine-grained motion strength control (1–10) for micro-loops.
- **Integrated Audio:** Native ambient soundtrack synthesis on the same canvas.

DIRECTING CAPABILITIES

- **Unified Canvas:** Seamless bridge from 2D edits to video
- **Motion Slider:** Fine-grained 1–10 motion scale
- **Integrated Audio:** Direct ambient soundtrack synthesis

Architecture:	Multimodal Canvas
Max Duration:	4s Ambient Loops
Camera Controls:	Motion Strength 1-10
Resolution:	720p / 1080p Upscale
Turnaround:	30-60 seconds
Access Model:	Daily Token Wallet

SELECTION HEURISTIC: Use Kling AI for humans and cycling action; use Runway for multi-camera splines; use Luma for rapid product spins.

TOOL SELECTION MATRIX

Motion Brush: Isolating Wheels and Water from Sensitive Faces

How to solve motion bleeding by painting discrete velocity vectors and freezing sensitive anatomy



Motion Brush Mask Mapping

Cyan: wheels (X = +4) • Amber: puddle spray (Y = +3) • Green: face & bag (Speed = 0)

BRUSH 1 • WHEEL ROTATION (VECTOR +X)

- **Masking Region:** Bicycle wheel rims, spokes, chain, and pedals.
- **Kinematic Vector:** Forward horizontal velocity vector (X = +4.0, Y = 0.0).
- **UI Settings:** Brush 1 • Sharp edge falloff • Linear translation.
- **Result:** Wheels rotate naturally without bending bicycle frame geometry.

BRUSH 2 • WATER HYDRODYNAMICS (VECTOR +Y)

- **Masking Region:** Wet cobblestone surface directly beneath tires.
- **Kinematic Vector:** Upward hydrodynamic splash vector (X = +1.5, Y = +3.0).
- **UI Settings:** Brush 2 • Soft radial gradient • Turbulent dispersion.
- **Result:** Water droplets disperse along authentic parabolic trajectories.

STATIC MASK • SPEED-0 FREEZE LOCK (VELOCITY = 0)

- **Masking Region:** Courier face, eyes, helmet, and Glovo delivery pack.
- **Kinematic Vector:** Velocity frozen strictly at zero (Speed = 0.0).
- **UI Settings:** Static Freeze Mask • 100% coordinate pin constraint.
- **Result:** 100% mathematical guarantee against facial distortion and logo morphing.

DIRECTING PROTOCOL: Always paint sensitive brand marks and human faces with Speed-0 Freeze Mask before setting background physics vectors!

LOCALIZED VECTOR CONTROL

5-Minute Coffee Break: Rest Your Eyes and Recharge

Strict rule: Zero chat tasks and zero questions during the pause. Step away from screens and recharge.

05:00

Coming up in Part 2: Multi-shot Storyboard Matrix (Glovo 4+4+3 case), Risk Governance Blast Radius, digital provenance via SynthID and C2PA, engineering decision framework, and homework assignment.

Rest your eyes: Look out the window at distant horizon • Zero chat discussions during break

The Storyboard Matrix: Assembling 3-Shot Commercials

Editing discipline: Rather than one unstable 15-second continuous take, generate 3 bounded shots and edit in timeline



SHOT 1 • ORDER PICKUP (4S)

Cafe, Barista, Kraft Bag

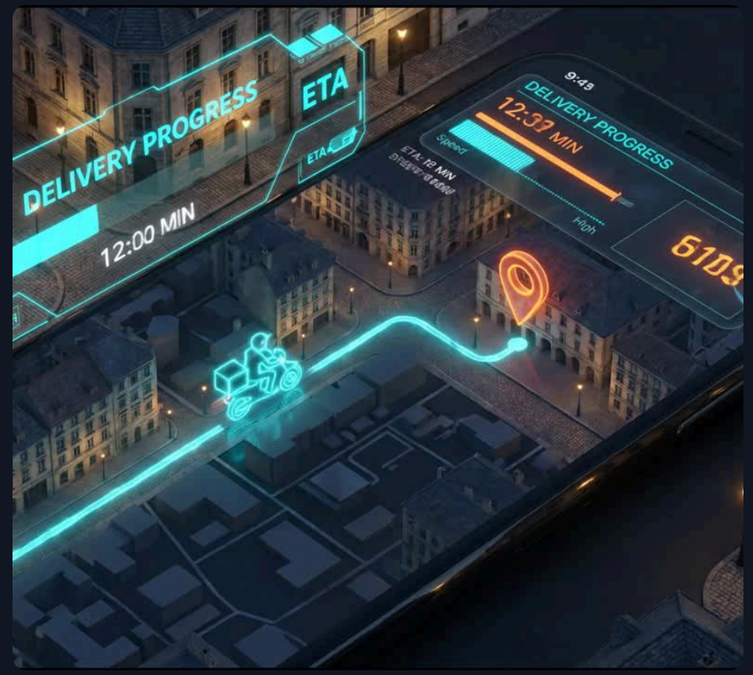
Framing:	Medium Shot (MS)
Timecode:	00:00 - 00:04 (4s)
Camera Vector:	Dolly-Out (Z = -2)
Foley Sound FX:	Cups, ambient jazz
Motion Guidance:	Two-hand bag lift



SHOT 2 • RAIN RIDE (4S)

Dynamic City Avenue Ride

Framing:	Tracking Close-Up (CU)
Timecode:	00:04 - 00:08 (4s)
Camera Vector:	Truck Left (X = +3)
Foley Sound FX:	Tire hiss, rain loop
Motion Guidance:	Wheel roll + splash



SHOT 3 • GPS ROUTE (3S)

3D Navigation Map on Smartphone

Framing:	Isometric Macro (ECU)
Timecode:	00:08 - 00:11 (3s)
Camera Vector:	Push-In (Z = +2)
Foley Sound FX:	Haptic chime, UI tap
Motion Guidance:	Route line animation

MONTAGE STANDARD (4+4+3): Three bounded shots assembled in CapCut or Premiere in 1 minute with 0.3s crossfade. 100% physical stability without long-take hallucinations!

TIMELINE ASSEMBLY

Safety Traffic Light & Digital Provenance: SynthID & C2PA

Where video AI is safe vs prohibited, and how cryptographic provenance manifests verify AI origin

GREEN • AUTONOMOUS GENERATION

- **Eligible Workloads:** Atmospheric B-roll, website background video loops, pitch animatics, moodboards.
- **Pre-Approved Models:** Kling Standard, Runway Gen-3, Luma Dream Machine.
- **Verification Protocol:** Automated brand filter scan; zero manual gatekeeping before internal review.
- **Legal Liability:** Zero commercial exposure; non-consumer facing internal material.

YELLOW • HUMAN-IN-THE-LOOP MANDATORY

- **Eligible Workloads:** Commercial client ads, e-commerce product packshots, social brand marketing, tutorials.
- **Mandatory QC Gate:** Frame-by-frame audit of anatomy (fingers, teeth), reflection physics, logo geometry.
- **Approval Sign-off:** Mandatory sign-off by Creative Director or Senior Brand QA before client delivery.
- **Risk Vector:** Brand damage and trust loss from mutated logos or uncanny valley defects.

RED • ABSOLUTE GENERATION BAN

- **Prohibited Workloads:** Breaking journalism, industrial safety SOPs, surgical procedures, likeness cloning.
- **Regulatory Exposure:** Criminal & civil liability under EU AI Act Articles 50 & 52.
- **Statutory Fines:** Up to €35M or 7% of total global annual turnover.
- **Enforcement Protocol:** Immediate PR rejection, automated DLP pipeline block, incident escalation to EPAM Legal.

GOOGLE SYNTHID • DEEP WATERMARKING

- **Core Mechanism:** Pseudo-random imperceptible perturbation embedded directly into the diffusion latent prior.
- **Resilience:** Survives lossy MP4/HEVC transcoding, resolution downsampling, cropping, and color grading.
- **Verification Flow:** Deep neural detector outputs calibrated confidence score (0.00–1.00) without requiring original.

C2PA CONTENT CREDENTIALS • CRYPTOGRAPHIC MANIFEST

- **Open Standard:** Coalition for Content Provenance and Authenticity (C2PA) standard (Adobe, Google, Microsoft, EPAM).
- **Cryptographic Binding:** SHA-256 frame hash bound to X.509 certificate and recorded in MP4 container metadata.
- **Lineage Traceability:** Captures foundation model ID, keyframe source SHA, generation timestamp, and operator identity.

ENTERPRISE QA & VERIFICATION TOOLCHAIN

- **Automated Manifest Scan:** CLI verification (c2patool) checking cryptographic lineage before publishing to CDN.
- **Visual QC Gate:** 5-point checklist: (1) finger grip contacts, (2) reflection physics, (3) logo text, (4) boundary jitter, (5) lip-sync.
- **Asset Lineage Archive:** Preserving master renders, generation seeds, and cryptographically signed manifests for regulatory audit.

REGULATORY COMPLIANCE: EU AI Act Article 52 mandates machine-readable watermarking and provenance disclosure for synthetic media.

COMPLIANCE PROTOCOL

Decision Tree: Selecting the Optimal Video Modality

4 diagnostic questions to save hours of GPU rendering and avoid budget drain

CRITERION 1: DOES THE SHOT REQUIRE SPEECH?

- **Diagnostic:** Does the scene require accurate lip synchronization with spoken voiceover?
- **Routing:** YES ➔ HeyGen / Hedra (Lip-Sync). Avoid full-body diffusion models!
- **Efficiency:** Saves 95% GPU compute and eliminates facial bone warping entirely.

HEYGEN / HEDRA (LIP-SYNC)

- **Architecture:** Facial landmark deformation driven by audio phoneme spectrograms.
- **Production Role:** Corporate avatars, onboarding tutorials, multilingual dubbing.
- **Key Advantage:** Sub-phoneme lip sync, zero body warping, 30s rendering speed.

CRITERION 2: SPECIFIC PRODUCT OR RECURRING CHARACTER?

- **Diagnostic:** Does the shot feature a recurring character or branded physical product?
- **Routing:** YES ➔ Strictly Image-to-Video (I2V). First lock keyframe in Chat UI ($0.04)!
- **Result:** Rejection rate plummets from 80% to under 15%; zero logo mutations.

KLING AI 1.5 / RUNWAY GEN-3

- **Architecture:** Spatiotemporal DiT conditioned on invariant boundary tensor at t=0.
- **Production Role:** Commercial brand ads, dynamic courier action, product promos.
- **Key Advantage:** 100% preservation of facial features, uniform colors, and brand logos.

CRITERION 3: ABSTRACT BACKGROUND OR NATURE B-ROLL?

- **Diagnostic:** Does the shot depict clouds, water currents, smoke, or ambient B-roll?
- **Routing:** YES ➔ Direct Text-to-Video (T2V) permitted using 6-slot prompt formula.
- **Rationale:** Natural fluid phenomena have no rigid geometric topological constraints.

LUMA DREAM MACHINE & LEONARDO

- **Architecture:** Rapid transformer diffusion with motion kinematics prompt prior.
- **Production Role:** Moodboard textures, website hero headers, ambient transitions.
- **Key Advantage:** 30-second turnaround, continuous fluid hydrodynamics, dynamic lighting.

CRITERION 4: HIGH-VOLUME PROGRAMMATIC PIPELINE?

- **Diagnostic:** Is this an automated programmatic batch pipeline (500+ catalog videos)?
- **Routing:** YES ➔ Cloud APIs (Google Veo 3.1 / Wan 2.2) with deterministic seeds.
- **Requirement:** Strict JSON schemas, async job queues, and metered cost auditing.

GOOGLE VEO 3.1 & WAN 2.2

- **Architecture:** Enterprise headless REST & gRPC endpoints with distributed GPU queues.
- **Production Role:** Automated programmatic video catalogs, personalized video scale.
- **Key Advantage:** Automated queue scheduling, enterprise SLAs, metered GPU cost accounting.

ENGINEERING PROTOCOL: Always prototype and validate motion kinetics in Web Studios before locking Cloud API pipelines.

DECISION BLUEPRINT

Homework: Animating a Hero Character or Product Packshot

Submit to genai-homeworks/L05/. Deadline: before next session. Collaborator: ihar_rubanovich@epam.com

TRACK A: E-COMMERCE PRODUCT PACKSHOT

360 Orbit Product Packshot

- **Creative Brief:** Commercial packshot for ceramic merchandise mug (d0_packshot.png from L04).
- **Kinematics:** Smooth Orbit 360 camera arc, rising turbulent steam convection, specular highlights.
- **Key Invariant:** Brand logo must remain 100% crisp and undistorted throughout the full 5s clip!
- **Recommended Tool:** Luma Dream Machine (Orbit mode) or Kling AI Image-to-Video.

PRODUCTION WORKFLOW

Disciplined 4-Step Production

- **1. Keyframe:** Select approved 2D asset from L04 or refine in Chat UI until defect-free ($0.04).
- **2. Prompt:** Formulate 6-slot director prompt specifying active verbs, camera motion, and negatives.
- **3. Masking:** In Kling/Runway, paint Motion Brush on active zones and Freeze Mask on logo/face.
- **4. Audit:** Synthesize identical keyframe in 2 distinct tools and complete markdown QA report.

TRACK B: GLOVO COURIER RIDE

Rainy Tracking Courier Ride

- **Creative Brief:** Dynamic courier transit sequence (shot2_bike_rain.png or custom L04 hero).
- **Kinematics:** Truck Left camera tracking parallel to cyclist speed, realistic puddle spray physics.
- **Key Invariant:** Speed-0 Freeze Mask over courier's face and delivery cube against logo drift.
- **Recommended Tool:** Kling AI 1.5 (Pro mode) or Runway Gen-3 Alpha.

REPOSITORY SUBMISSION STRUCTURE

- **videos/:** 2 MP4 video files comparing generation outputs across 2 tools (Kling vs Runway/Luma).
- **video_spec.md:** 6-slot director prompts, camera motion vectors, seeds, duration, settings.
- **video_qa.md:** Structured comparative defect audit table comparing tools across 5 dimensions.

```
genai-homeworks/L05/
├── videos/
│   ├── track_a_kling.mp4     [5s, 1080p, Orbit 360]
│   └── track_a_runway.mp4    [5s, 1080p, Orbit 360]
├── video_spec.md               [Prompts, seeds, vectors]
└── video_qa.md                 [Defect audit matrix]
```

GRADING RUBRIC & ACCEPTANCE CRITERIA

- **40 pts — Keyframe Fidelity:** Zero identity and branding drift between L04 asset and L05 video.
- **30 pts — Kinematic Camera Execution:** Precise execution of requested trajectory (Orbit 360 or Truck Left).
- **20 pts — Physical Realism:** Authentic fluid dynamics (steam, water splash) without mesh meltdown.
- **10 pts — Rigorous Comparative QA:** Grounded technical evaluation matrix in video_qa.md.

```
git checkout -b homework/L05-<your-name>
git add videos/ video_spec.md video_qa.md
git commit -m "feat(L05): multi-tool video synthesis benchmark"
git push origin homework/L05-<your-name> && gh pr create
```

Notice: Zero code required! All video synthesis is performed via Web UI and Chat UI. Submit PR to genai-homeworks/L05/ before the next session.

Four Golden Rules & Lesson 06 Teaser: Speech & Audio AI

All our clips today were completely silent. In our next session, we bring them alive with voice and music!

1. The Keyframe Axiom (I2V)

- **Principle:** Commercial video begins with an approved static 2D keyframe from Chat UI (~$0.04).
- **Mechanics:** Initial tensor locking at t=0 collapses latent entropy by >80% and eliminates drift.

2. The Delta Principle (dx/dt)

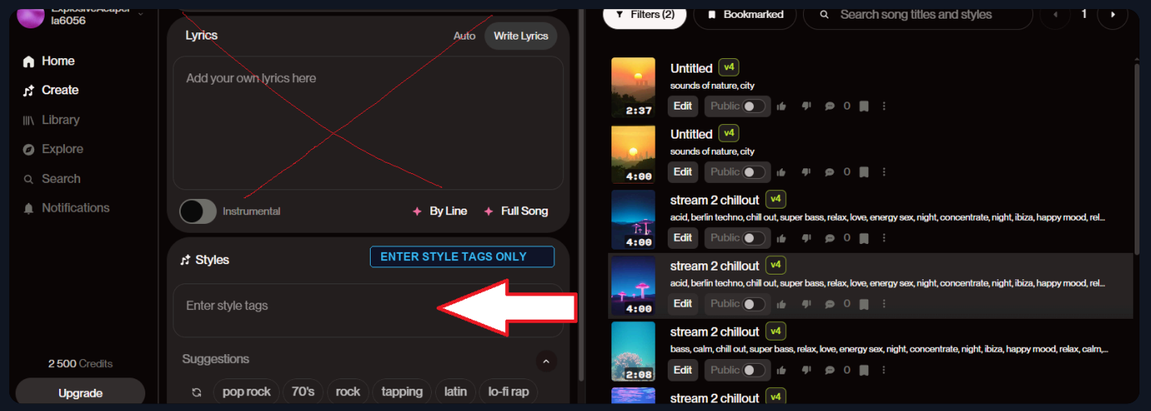
- **Principle:** Prompts direct strictly kinematics and temporal velocity deltas over time.
- **Rule:** Never re-describe visual entities, costumes, or backgrounds already frozen in the initial keyframe.

3. The Montage Discipline (4+4+3)

- **Principle:** Assemble 3 short bounded shots (3–5s each) via timeline editing with 0.3s crossfades.
- **Rejection:** Never attempt an unstable 15s continuous take in a single video generation prompt.

4. The Budget Economy

- **Principle:** Disciplined I2V costs $2–$5 per finished commercial video using locked anchors.
- **Trap:** Naive Text-to-Video burns $50–$100 on stochastic rerolls with an 80%+ rejection rate.



L06 TEASER • SUNO AI INTERFACE Soundtracks & Foley Audio

NEXT SESSION • L06: SPEECH & AUDIO AI

- **ElevenLabs:** voice cloning and emotional intonation control.
- **Suno & Udio:** royalty-free music synthesis across genres and moods.
- **Foley Sound FX:** rain noise, tire hiss, and coffee shop ambient clatter.
- **Lip-Sync:** exact phoneme lip-syncing for our animated courier!

OPEN MIC & Q&A: Share your video generation questions in chat or unmute to discuss!

SEE YOU IN LESSON 06